Shortest Common Supersequence | Introduction & SCS Length

The shortest common supersequence (SCS) is the problem of finding the shortest supersequence Z of given sequences X and Y such that both X & Y are subsequences of Z.

```
For example, consider the two
following sequences X and Y

X: ABCBDAB
Y: BDCABA


The length of SCS is 9
SCS are ABCBDCABA, ABDCABDAB and
ABDCBDABA
```

The SCS problem has an optimal substructure. That means the problem can be broken down into smaller, simple

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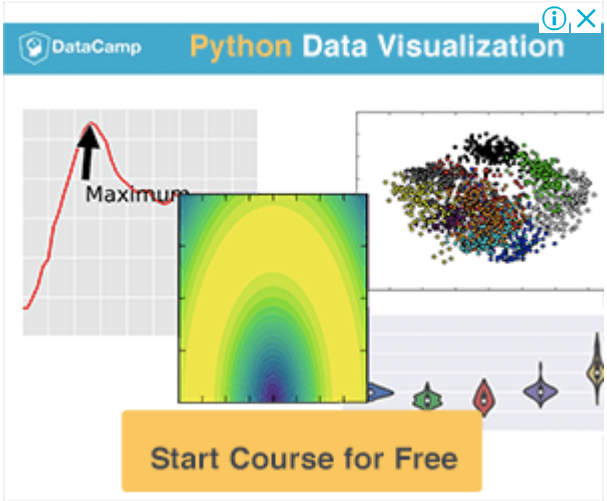

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"subproblems", which can be broken down into yet simpler subproblems, and so on, until, finally, the solution becomes trivial.

1. Let us consider two sequences X and Y of length m and n respectively that both end in the same element.

To find their SCS, we shorten each sequence by removing the last element, find the SCS of the shortened sequences, and to that SCS append the removed element. So we can say that

```
SCS(X[1..m], Y[1..n]) = SCS(X[1..m-1], Y[1..n-1]) + X[m]   if X[m] = Y[n]
```

For example,

```
SCS('ABCBDA', 'BDCABA') = SCS('ABCBD', 'BDCAB') + 'A'
SCS('ABCBDAB', 'BDCAB') = SCS('ABCBDA', 'BDCA') + 'B'
```

2. Now suppose that the two sequences do not end in the same element.

Then the SCS of X and Y is the **shorter** of the two sequences `SCS(X[1..m-1], Y[1..n]) + X[m]` and `SCS(X[1..m], Y[1..n-1]) + Y[n]`. To understand this property, let's consider the two following sequences

```
X: ABCBDAB (n elements)
Y: BDCABA  (m elements)
```

The SCS of these two sequences either ends with a B (the last element of sequence X) or with A (the last element of sequence Y).

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Case 1: If SCS ends with a B, then we add B from sequence X to SCS and the problem reduces to `SCS(X[1..m-1], Y[1..n]) +` *`X[m]`*.

Case 2: If SCS does not end with a B, that means it ends with A, then we add A from sequence Y to SCS and the problem reduces to `SCS(X[1..m], Y[1..n-1]) +` *`Y[n]`*.

For example,

```
SCS('ABCBD', 'BDCAB') =
minimum (SCS('ABCB', 'BDCAB') + 'D',
                                SCS('
ABCBD', 'BDCA') + 'B')
```

Below solution finds the length of shortest common supersequence of sequences `X[0..m-1]` and `Y[0..n-1]` recursively by using optimal substructure property of SCS problem.

C++

```
#include <bits/stdc++.h>
using namespace std;

// Function to find length of sho
// sequences X[0..m-1] and Y[0..n
int SCSLength(string X, string Y,
{
    // if we have reached the end
    // length of other sequence
    if (m == 0 || n == 0)
        return n + m;

    // if last character of X and
    if (X[m - 1] == Y[n - 1])
        return SCSLength(X, Y, m
    else
    // else if last character of
        return min(SCSLength(X, Y
                   SCSLength(X,
}

// main function
```

```
int main()
{
    string X = "ABCBDAB", Y = "BD
    int m = X.length(), n = Y.len

    cout << "The length of shorte
            << SCSLength(X, Y, m,

    return 0;
}
```

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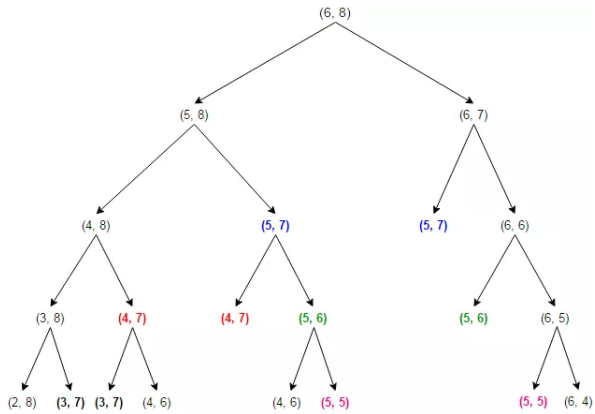
Output:

```
The length of shortest Common
supersequence is 9
```

The worst case time complexity of above solution is $O(2^{(m+n)})$ and auxiliary space used by the program is $O(1)$. The worst case happens when there is no common character present in X and Y (i.e. SCS length is $(m+n)$) and each recursive call will end up in two recursive calls.

The SCS problem exhibits overlapping subproblems. Let us consider recursion tree for two sequences of length 6 and 8 whose SCS length is $(6+8=14)$. As we can see, the same sub-problems (highlighted in same color) are getting computed again and again –



The problems having optimal substructure & overlapping subproblems can be solved by dynamic programming, in which subproblem solutions are memoized rather than computed again and again. Below is *Memo*ized version that follows the top-down approach, as we first break the problem into subproblems and then calculate and store values.

C++

```
#include <bits/stdc++.h>
using namespace std;

// Function to find length of sho
// sequences X[0..m-1] and Y[0..n
int SCSLength(string X, string Y,
{
    // if we have reached the end
    // length of other sequence
    if (m == 0 || n == 0)
        return n + m;

    // construct a unique map key
    string key = to_string(m) + "

    // if sub-problem is seen for
    // store its result in a map
    if (lookup.find(key) == looku
    {
        // if last character of X
        if (X[m - 1] == Y[n - 1])
            lookup[key] = SCSLeng

        else

```

```
        // else if last character
        lookup[key] = min (SCSLen
                                SCS
    }

    // return the subproblem solu
    return lookup[key];
}

// main function
int main()
{
    string X = "ABCBDAB", Y = "BD
    int m = X.length(), n = Y.len

    // create a map to store solu
    // we can also use array inst
    unordered_map<string, int> lo

    cout << "The length of shorte
            << SCSLength(X, Y, m,

    return 0;
}
```

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Output:

```
The length of shortest Common
supersequence is 9
```

The time complexity of above solution is `O(mn)` and auxiliary space used by the program is `O(mn)`.

We can also solve this problem in bottom-up manner. In the bottom-up approach, we calculate the smaller values of `SCS(i, j)` first, then build larger values from them.

```
            |
j
 (if i == 0)
            |
i
 (if j == 0)
SCS[i][j] = | SCS[i-1][j-1] +
1                      (if X[i-1] ==
Y[j-1])
            | min (SCS[i-1][j] + 1,
SCS[i][j-1] + 1) (if X[i-1] != Y[j-1])
```

C++

```cpp
#include <bits/stdc++.h>
using namespace std;

// Function to find length of sho
// sequences X[0..m-1] and Y[0..n
int SCSLength(string X, string Y)
{
    int m = X.length(), n = Y.len

    // lookup table stores soluti
    // i.e. lookup[i][j] stores t
    // X[0..i-1] and Y[0..j-1]
    int lookup[m + 1][n + 1];

    // initalize first column of
    for (int i = 0; i <= m; i++)
        lookup[i][0] = i;

    // initalize first row of the
    for (int j = 0; j <= n; j++)
        lookup[0][j] = j;

    // fill the lookup table in b
    for (int i = 1; i <= m; i++)
    {
        for (int j = 1; j <= n; j
        {
            // if current charact
            if (X[i - 1] == Y[j -
                lookup[i][j] = lo

            // else if current ch
            else
                lookup[i][j] = mi
                                1
        }
```

```
    }

    // SCS will be last entry in
    return lookup[m][n];
}

// main function
int main()
{
    string X = "ABCBDAB", Y = "BD

    cout << "The length of shorte
            << SCSLength(X, Y);

    return 0;
}
```

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Output:

```
The length of shortest Common
supersequence is 9
```

The time complexity of above solution is `O(mn)` and auxiliary space used by the program is `O(mn)`. The space complexity of above solution can be improved to `O(n)` as calculating SCS of a row of the SCS table requires only the solutions to the current row and the previous row.

Read: Shortest Common Supersequence (Using LCS)

Exercise:

1. Extend the solution for Printing Shortest Common Supersequence.
2. Write space optimized code for iterative

version.

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Prabhat

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